

UMME FARHANA SUMAIYA

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PROFESSIONAL SUMMARY

Final-year ICT student at BUP with hands-on experience building production-grade backend systems and AI-integrated applications. Currently interning at Brainstation23, where I have built data ingestion pipelines, ML-integrated APIs, and geospatial systems for real-world Dhaka-based projects. Experienced in Python, PostgreSQL, and FastAPI. Conducting undergraduate thesis on heart arrhythmia detection using CNN and Vision Transformer. Led IEEE Computer Society BUP Student Branch as Chairperson and actively contribute to technical community building.

Core Skills: Python, FastAPI, PostgreSQL, Machine Learning, Data Pipelines, REST APIs

EDUCATION

Bangladesh University of Professionals

Bachelor of Science in Information and Communication Engineering

CGPA 3.36/4.00 (up to 7th semester)

Mirpur Cantonment, Dhaka

Jul. 2022 – Present

BAF Shaheen College, Jashore

Higher Secondary School Certificate (GPA 5.00/5.00)

Jashore

2020

WORK EXPERIENCE

Brainstation23 PLC

Intern (2-month Industrial Attachment)

Dhaka, Bangladesh

May 2026 – Present

Project 1: Real-time AI-Powered Traffic & Accident Prediction System, Dhaka

Built the end-to-end data ingestion pipeline for a 6-person intern project predicting traffic congestion and accident risk across 50+ Dhaka road segments.

- Designed a Celery Beat scheduler polling Google Maps Distance Matrix API every 10 minutes, storing time-series data in PostgreSQL + PostGIS
- Engineered Dhaka-specific features: hartal days, Ramadan patterns, Friday prayer hours, and real-time rain intensity from OpenWeatherMap API
- Processed Dhaka's road network via OSMnx, enriching segments with road width and waterlogging classification for accident risk modeling
- Built an automated gap-detection and linear interpolation pipeline, combining 2,500 real-world readings with domain-rule-based synthetic data, delivering 5,000 labeled rows for ML model training

Tech: Python, Celery, Redis, PostgreSQL, PostGIS, OSMnx, SQLAlchemy

Project 2: Smartest Tech Deals — Live Phone Price Comparison Platform, Bangladesh

Built the ML pipeline, admin dashboard, and search/AI intelligence layer for a 6-person team project aggregating real-time phone prices across Bangladeshi e-commerce platforms.

- Clustered search queries using DBSCAN + TF-IDF (char n-gram) with automatic eps tuning via the Elbow Method, identifying trending product searches in real time
- Built hourly spike detection using Isolation Forest with percentage-change fallback, scoring search volume anomalies across all queries
- Designed search-regret and dead-query detection using fuzzy string matching, surfacing underserved searches and product catalog gaps
- Built a Next.js admin dashboard with metrics cards, Recharts-based trending/intent visualization, scraper health monitoring, and protected routes via middleware
- Integrated the ML pipeline with a FastAPI backend via authenticated REST endpoints, scheduled hourly via Docker cron
- Implemented fuzzy-match search autocomplete with typo correction and trending-query fallback to eliminate empty search results
- Integrated an LLM-powered chatbot into client-facing pages for natural-language product assistance
- Resolved cross-team data and integration issues across backend, database, and DevOps layers during live integration testing

Tech: Python, scikit-learn, pandas, asyncpg, RapidFuzz, PostgreSQL, MongoDB, Next.js, Recharts, TypeScript

PROJECTS

Brain Tumor Detection Web Application

Dec 2025

Python, Deep Learning, Explainable AI

- Developed a web application detecting brain tumors from MRI images using deep learning, with bounding-box localization of the affected region
- Added an Explainable AI (XAI) module providing visual explanations for the model's predictions

Book Classifier App

Nov 2025

Android Studio, TensorFlow Lite

- Built an Android app classifying books from cover images using an on-device TensorFlow Lite model trained via Teachable Machine, with camera and gallery input support

Other Projects: PlasticTrack+ (Python/Tkinter/SQLite desktop app for plastic consumption tracking with automated PDF reporting) · Prochestha (PHP/MySQL habit-tracking web app) · Ahoron (PHP/MySQL multi-role university club platform) · UCamLite (Java/MySQL university management system)

RESEARCH & LEADERSHIP EXPERIENCE

Undergraduate Thesis Researcher | CNN, Vision Transformer (ViT)

Jul. 2025 – Present

Bangladesh University of Professionals — Dhaka, Bangladesh

- Conducting undergraduate thesis research on heart arrhythmia detection using deep learning, implementing and comparing CNN and ViT architectures for medical signal classification

Chairperson

Oct. 2025 – Present

IEEE Computer Society, BUP Student Branch Chapter — Dhaka, Bangladesh

- Lead organizational activities and coordinate technical events, workshops, and seminars; organized IEEE BranchFest 2025 and a day-long Web/Git Basics workshop

Vice President (Communication)

Oct. 2025 – Present

BUP Robotics Club — Dhaka, Bangladesh

- Manage internal and external communications for club activities, announcements, and collaborations

Management Trainee / Intern

Mar. 2025 – Dec. 2025

YSSE, Content Writing Department — Dhaka, Bangladesh

- Wrote and reviewed blog content, guided four interns on writing quality, and managed publishing for the YSSE Blog
- Recognized as Management Trainee of the Month (Nov 2025) and Intern of the Month (May 2025); achieved Gold certification

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, Java, SQL, HTML, CSS

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, Next.js, Tailwind CSS, Scikit-learn, Pandas, NumPy, TensorFlow Lite

Databases: PostgreSQL, PostGIS, MySQL, MongoDB, SQLite, Firebase

Tools & DevOps: Redis, Celery, Git, Postman, SQLAlchemy, asyncpg, Docker

APIs & Geospatial: Google Maps API, OpenWeatherMap API, OSMnx

ML & AI: Scikit-learn, Deep Learning, Explainable AI (XAI), DBSCAN, Isolation Forest, CNN, Vision Transformer

CERTIFICATIONS

- CS50's Introduction to Programming with Python ([CS50P](#))
- [AWS Cloud Technical Essentials](#) -- GP Academy
- [Full Stack Development with MERN](#) -- GP Academy